



Servidor VoIP Asterisk

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Servidor VoIP Asterisk

1. Instalación de Asterisk

Para poder instalar el servicio de Asterisk tendremos que actualizar el sistema con el **comando** `sudo apt update` y `sudo apt upgrade -y`

Cuando ya lo tengamos actualizado ejecutaremos el **comando** `wget` <https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-21-current.tar.gz>

Creamos una carpeta llamada asterisk y otra llamada src con el **comando** `mkdir (nombre de la carpeta)`

Cuando tengamos el archivo de instalación lo moveremos a la carpeta de src con el **comando** `sudo mv` [asterisk-21-current.tar.gz](https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-21-current.tar.gz) /asterisk/src

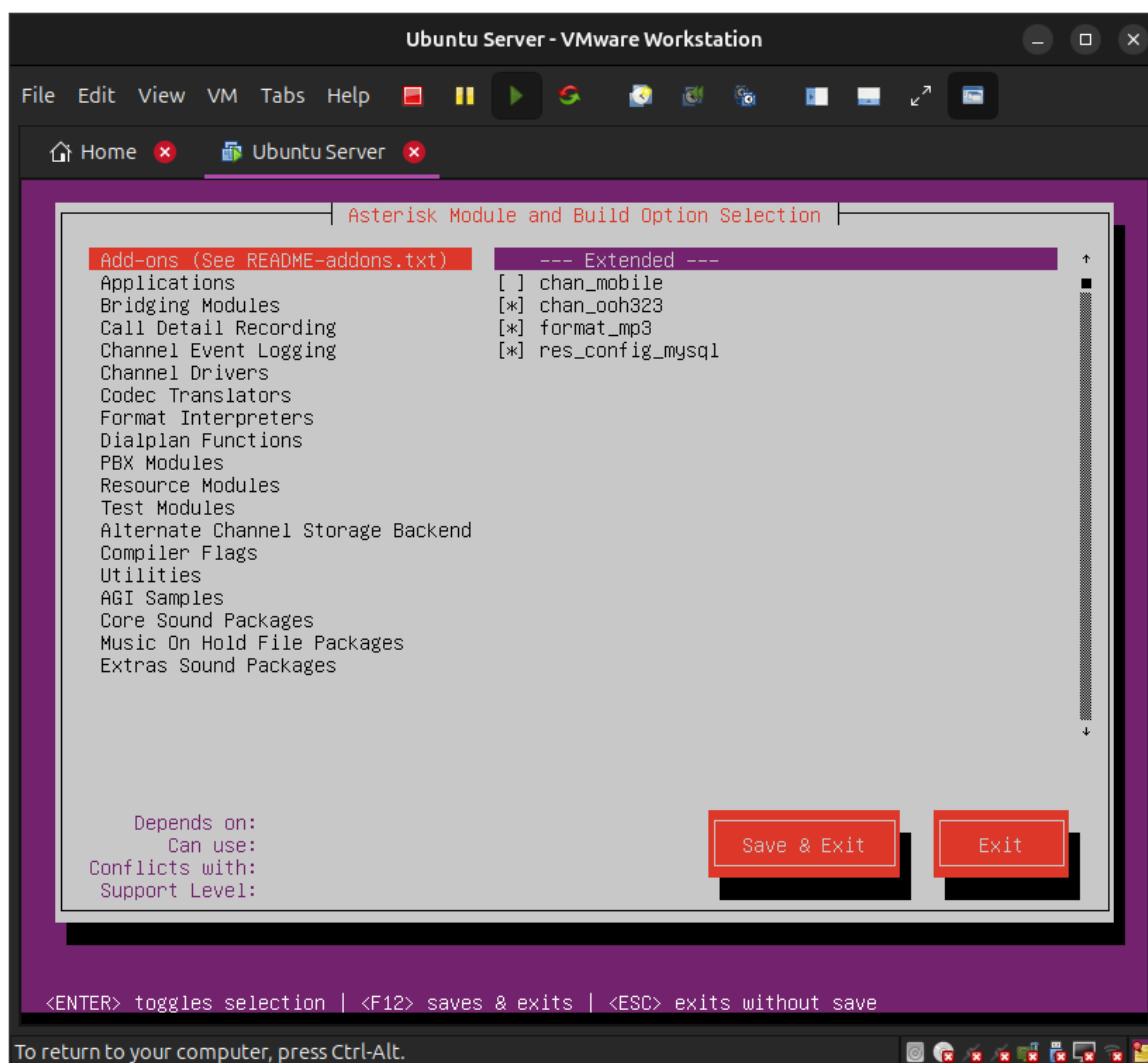
Ejecutaremos el **comando** `sudo tar -xvzf` [asterisk-21-current.tar.gz](https://downloads.asterisk.org/pub/telephony/asterisk/asterisk-21-current.tar.gz)

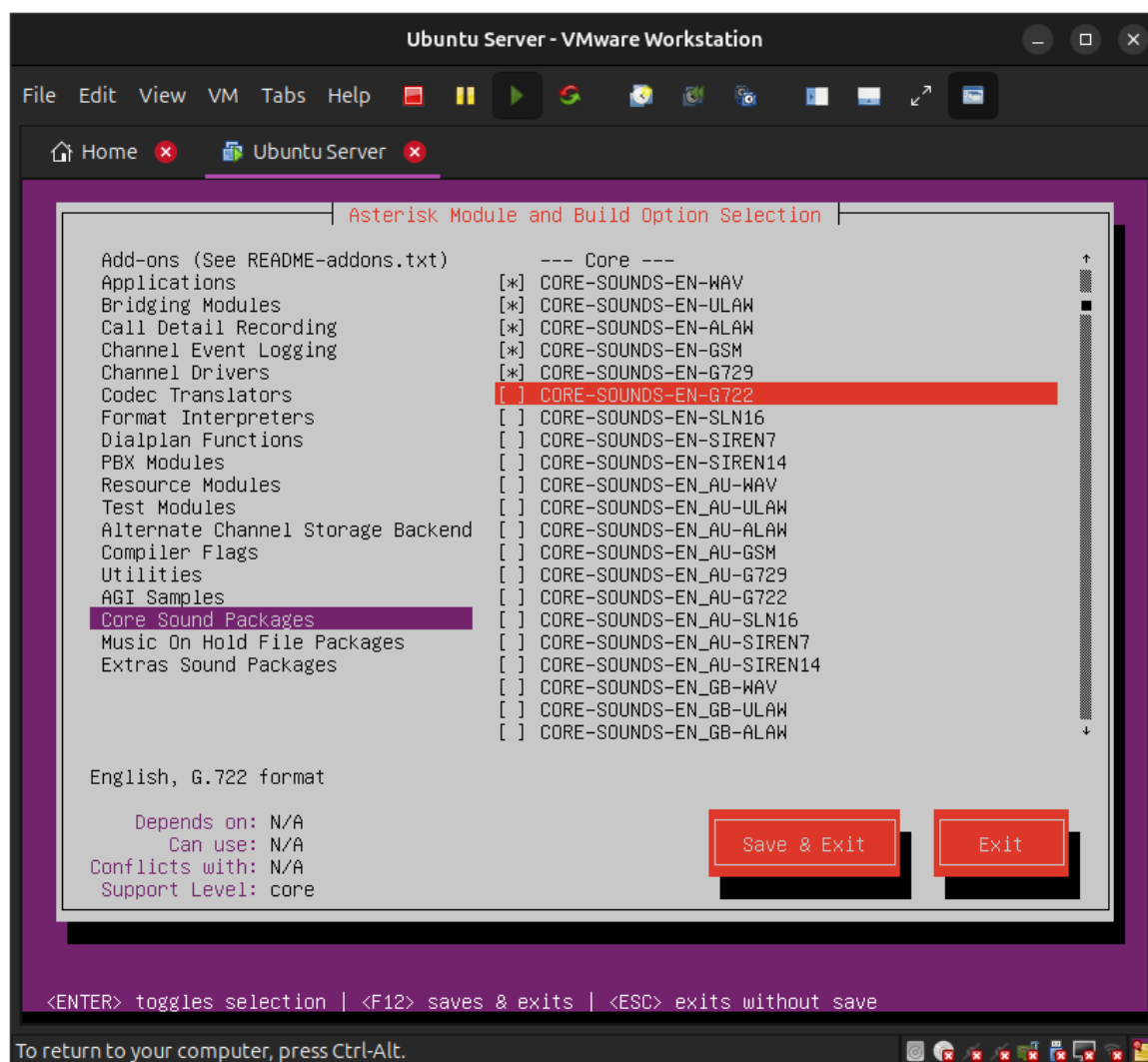
Nos meteremos a la carpeta con el **comando** `cd asterisk-21-current` y ejecutaremos el **comando** `contrib/scripts/get_mp3_source.sh` para ejecutar ese archivo

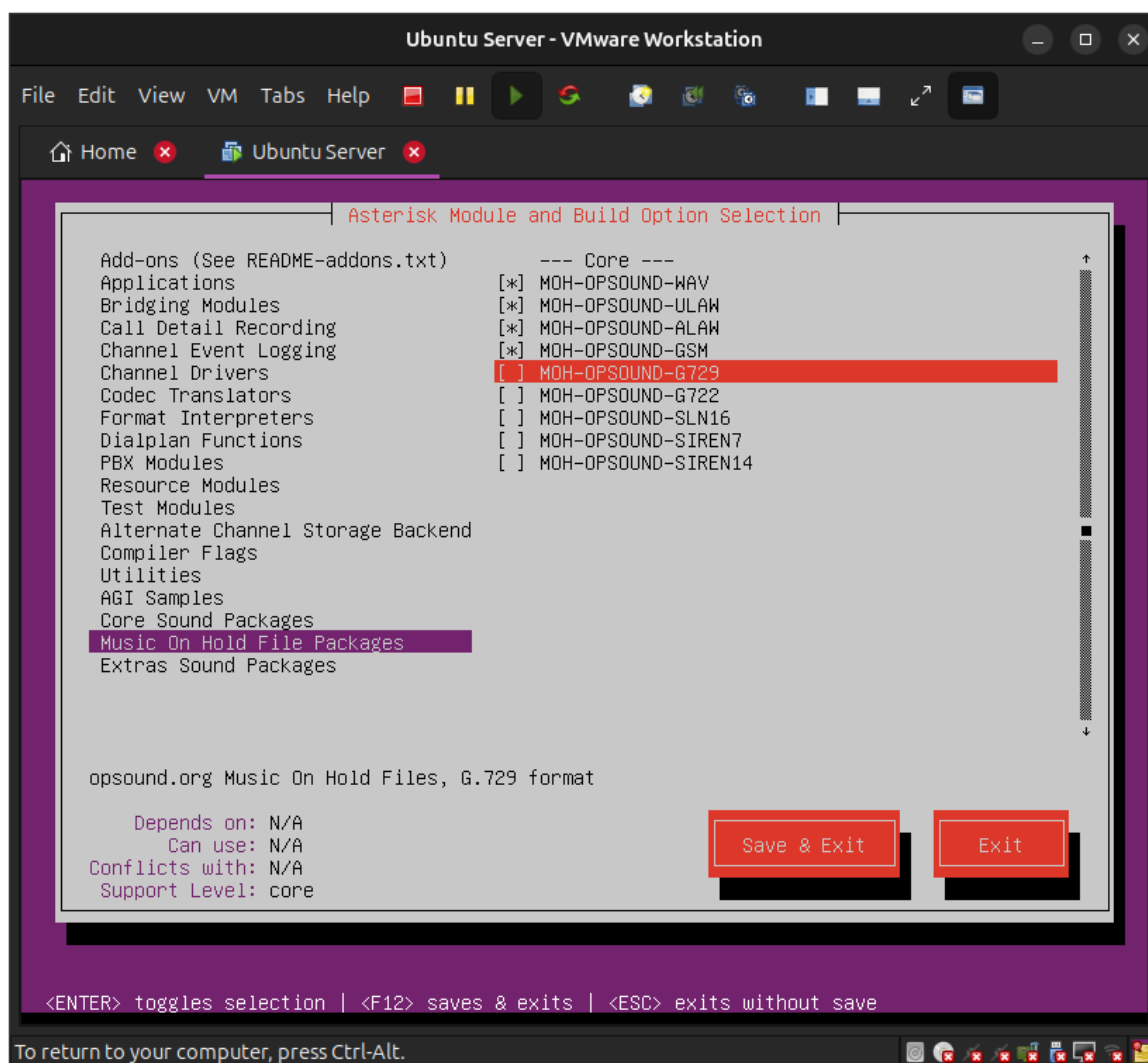
Ahora instalaremos las dependencias con el **comando** `sudo contrib/scripts/install_prereq install`

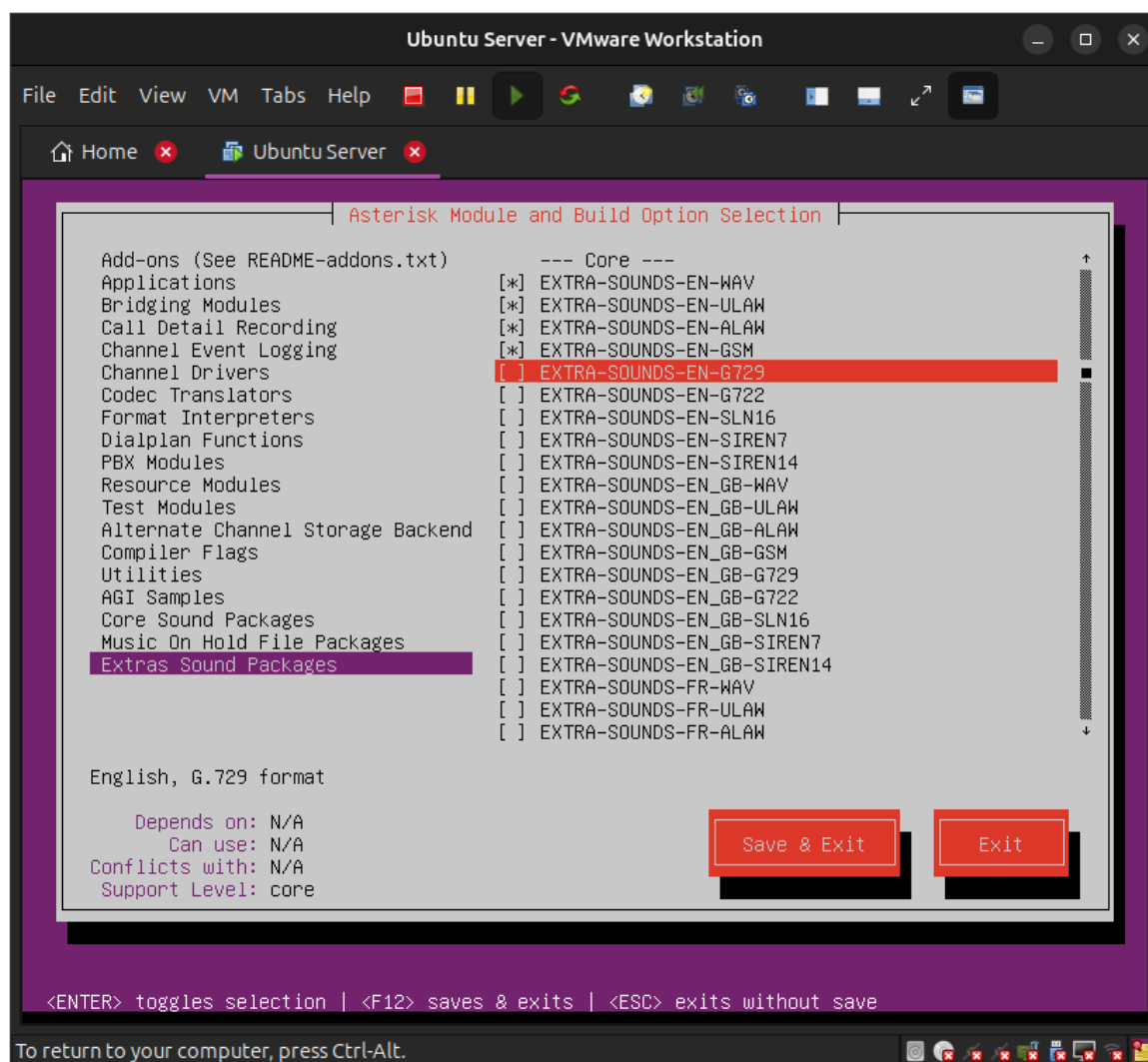
Configuraremos asterisk con el **comando** `sudo ./configure`

Ahora instalaremos los siguientes módulos con el **comando** `sudo make menuselect`









Ahora haremos la construcción de asterisk con el **comando sudo make**, ejecutaremos el **comando sudo make install** para la instalación, el **comando sudo make samples** y **sudo make config** y ejecutaremos el **comando ldconfig** para ver que se haya creado correctamente

Ubuntu Server - VMware Workstation

File Edit View VM Tabs Help

Home x Ubuntu Server x

```
[pjproject] Compiling libpj-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjlib-util-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsua-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-ua-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-simple-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjsip-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-codec-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-videodev-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjmedia-audiodev-x86_64-unknown-linux-gnu.a
[pjproject] Compiling libpjnath-x86_64-unknown-linux-gnu.a
[pjproject] Generating symbols
[CC] chan_audiosocket.c -> chan_audiosocket.o
[LD] chan_audiosocket.o -> chan_audiosocket.so
[CC] chan_bridge_media.c -> chan_bridge_media.o
[LD] chan_bridge_media.o -> chan_bridge_media.so
[CC] chan_console.c -> chan_console.o
[LD] chan_console.o -> chan_console.so
[CC] chan_iax2.c -> chan_iax2.o
[CC] iax2/codecs_pref.c -> iax2/codecs_pref.o
[CC] iax2/firmware.c -> iax2/firmware.o
[CC] iax2/format_compatibility.c -> iax2/format_compatibility.o
[CC] iax2/netsock.c -> iax2/netsock.o
[CC] iax2/parser.c -> iax2/parser.o
[CC] iax2/provision.c -> iax2/provision.o
[LD] chan_iax2.o iax2/codecs_pref.o iax2/firmware.o iax2/format_compatibility.o iax2/netsock.o iax2/parser.o iax2/provision.o -> chan_iax2.so
[CC] chan_motif.c -> chan_motif.o
[LD] chan_motif.o -> chan_motif.so
[CC] chan_pjsip.c -> chan_pjsip.o
[CC] pjsip/cli_commands.c -> pjsip/cli_commands.o
[CC] pjsip/dialplan_functions.c -> pjsip/dialplan_functions.o
[LD] chan_pjsip.o pjsip/cli_commands.o pjsip/dialplan_functions.o -> chan_pjsip.so
[CC] chan_rtp.c -> chan_rtp.o
[LD] chan_rtp.o -> chan_rtp.so
[CC] chan_unistim.c -> chan_unistim.o
```

To direct input to this VM, click inside or press Ctrl+G.



```
Ubuntu Server - VMware Workstation
File Edit View VM Tabs Help
Home x Ubuntu Server x
[CC] ooh323c/src/printHandler.c -> ooh323c/src/printHandler.o
[CC] ooh323c/src/rcttype.c -> ooh323c/src/rcttype.o
[CC] ooh323c/src/ooTimer.c -> ooh323c/src/ooTimer.o
[CC] ooh323c/src/h323/H235-SECURITY-MESSAGESDec.c -> ooh323c/src/h323/H235-SECURITY-MESSAGESDec.o
[CC] ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.c -> ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.o
[CC] ooh323c/src/h323/H323-MESSAGES.c -> ooh323c/src/h323/H323-MESSAGES.o
[CC] ooh323c/src/h323/H323-MESSAGESDec.c -> ooh323c/src/h323/H323-MESSAGESDec.o
[CC] ooh323c/src/h323/H323-MESSAGESEnc.c -> ooh323c/src/h323/H323-MESSAGESEnc.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.o
[CC] ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.c -> ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.o
[CC] ooh323cDriver.c -> ooh323cDriver.o
[LD] chan_ooh323.o ooh323c/src/ooCmdChannel.o ooh323c/src/ooLogChan.o ooh323c/src/ooUtils.o ooh323c/src/ooGKClient.o ooh323c/src/context.o ooh323c/src/ooDateTime.o ooh323c/src/decode.o ooh323c/src/dlist.o ooh323c/src/encode.o ooh323c/src/errmgmt.o ooh323c/src/memheap.o ooh323c/src/ootrace.o ooh323c/src/oochannels.o ooh323c/src/ooh245.o ooh323c/src/ooports.o ooh323c/src/ooq931.o ooh323c/src/ooCapability.o ooh323c/src/ooSocket.o ooh323c/src/perutil.o ooh323c/src/eventHandler.o ooh323c/src/ooCall.o ooh323c/src/ooStackCmds.o ooh323c/src/ooh323.o ooh323c/src/ooh323ep.o ooh323c/src/printHandler.o ooh323c/src/rcttype.o ooh323c/src/ooTimer.o ooh323c/src/h323/H235-SECURITY-MESSAGESDec.o ooh323c/src/h323/H235-SECURITY-MESSAGESEnc.o ooh323c/src/h323/H323-MESSAGES.o ooh323c/src/h323/H323-MESSAGESDec.o ooh323c/src/h323/H323-MESSAGESEnc.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROL.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLDec.o ooh323c/src/h323/MULTIMEDIA-SYSTEM-CONTROLEnc.o ooh323cDriver.o -> chan_ooh323.so
[CC] res_config_mysql.c -> res_config_mysql.o
[LD] res_config_mysql.o -> res_config_mysql.so
Building Documentation For: channels pbx apps codecs formats cdr cel bridges funcs tests main res ad
dons
+----- Asterisk Build Complete -----+
+ Asterisk has successfully been built, and +
+ can be installed by running: +
+ +
+ make install +
+-----+
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo make install make samples make configure ld
config_

To return to your computer, press Ctrl-Alt.
```



```
Ubuntu Server - VMware Workstation
File Edit View VM Tabs Help
Home Ubuntu Server

asterisk-moh-opsound-gsm 100X[=====] 1.70M 61.4KB/s in 29s
2026-01-25 19:58:25 (59.8 KB/s) - 'asterisk-moh-opsound-gsm-2.03.tar.gz' saved [1777967/1777967]
make[1]: Leaving directory '/home/ivan/asterisk/src/asterisk-21.12.0/sounds'
find rest-api -name "*.json" | while read x; do \
    /usr/bin/install -c -m 644 $x "/var/lib/asterisk/rest-api" ; \
done
----- Asterisk Installation Complete -----
+
+ YOU MUST READ THE SECURITY DOCUMENT
+
+ Asterisk has successfully been installed.
+ If you would like to install the sample
+ configuration files (overwriting any
+ existing config files), run:
+
+ For generic reference documentation:
+ make samples
+
+ For a sample basic PBX:
+ make basic-pbx
+
+ ----- or -----
+
+ You can go ahead and install the asterisk
+ program documentation now or later run:
+
+ make progdocs
+
+ **Note** This requires that you have
+ doxygen installed on your local system
+
make: *** No rule to make target 'make'. Stop.
ivan@ivan-laptop:~/asterisk/src/asterisk-21.12.0$ _
```

To direct input to this VM, click inside or press Ctrl+G.



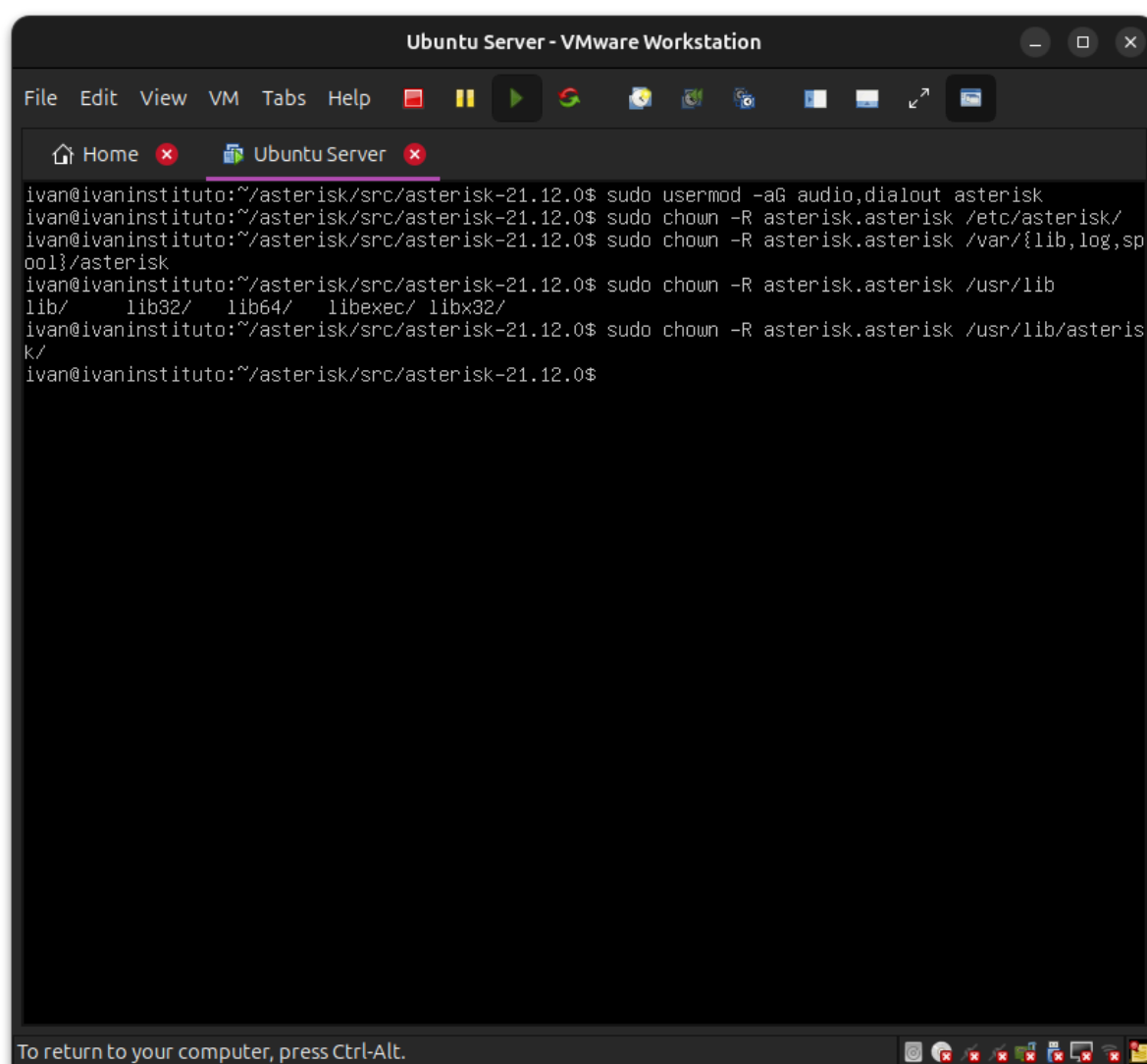
2. Creación de usuarios

The screenshot shows a terminal window titled "Ubuntu Server - VMware Workstation". The terminal has two tabs: "Home" and "Ubuntu Server". The command prompt is `ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$`. The user enters `sudo groupadd asterisk`, followed by a password prompt `[sudo] password for ivan:`. Then, the user enters `sudo useradd -r -d /var/lib/asterisk/ -g asterisk asterisk_`. The terminal output shows the command being executed. At the bottom of the window, there is a message: "To return to your computer, press Ctrl-Alt." and a taskbar with several application icons.

```
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo groupadd asterisk
[sudo] password for ivan:
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo useradd -r -d /var/lib/asterisk/ -g asterisk
k asterisk_
```



```
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo usermod -aG audio,dialout asterisk
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$
```



```
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo usermod -aG audio,dialout asterisk
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo chown -R asterisk.asterisk /etc/asterisk/
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo chown -R asterisk.asterisk /var/{lib,log,spool}/asterisk
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo chown -R asterisk.asterisk /usr/lib
lib/ lib32/ lib64/ libexec/ libx32/
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$ sudo chown -R asterisk.asterisk /usr/lib/asterisk/
ivan@ivaninstituto:~/asterisk/src/asterisk-21.12.0$
```

Ahora crearemos los usuarios para asterisk con el comando **sudo groupadd asterisk** para añadir el **grupo** asterisk. El comando **sudo useradd -r -d /var/lib/asterisk -g asterisk asterisk** para crear el **usuario** de asterisk

Añadiremos el usuario de audio y dialout al grupo de asterisk con el **comando sudo usermod -aG audio,dialout asterisk**

Cambiaremos la propiedad del directorio de configuración de asterisk con el **comando sudo chown -R asterisk.asterisk /etc/asterisk**



```
sudo chown -R asterisk.asterisk /var/{lib,log,spool}/asterisk
```

```
sudo chown -R asterisk.asterisk /usr/lib/asterisk
```

3. Configuración Asterisk

```
Ubuntu Cliente - VMware Workstation
File Edit View VM Tabs Help
GNU nano 6.2 /etc/default/asterisk
# Startup configuration for the Asterisk daemon

# Uncomment the following and set them to the user/groups that you
# want to run Asterisk as. NOTE: this requires substantial work to
# be sure that Asterisk's environment has permission to write the
# files required for its operation, including logs, its comm
# asterisk database, etc.
AST_USER='asterisk'
AST_GROUP='asterisk'

# If you DON'T want Asterisk to start up with terminal colors, comment
# this out.
COLOR=yes

# If you want Asterisk to run with a non-default configuration file,
# uncomment the following option, and set the value appropriately.
#ALTCONF=/etc/asterisk/asterisk.conf

# In the case of a crash, Asterisk may create a core file. Uncomment
# if you want this behavior.
#COREDUMP=yes

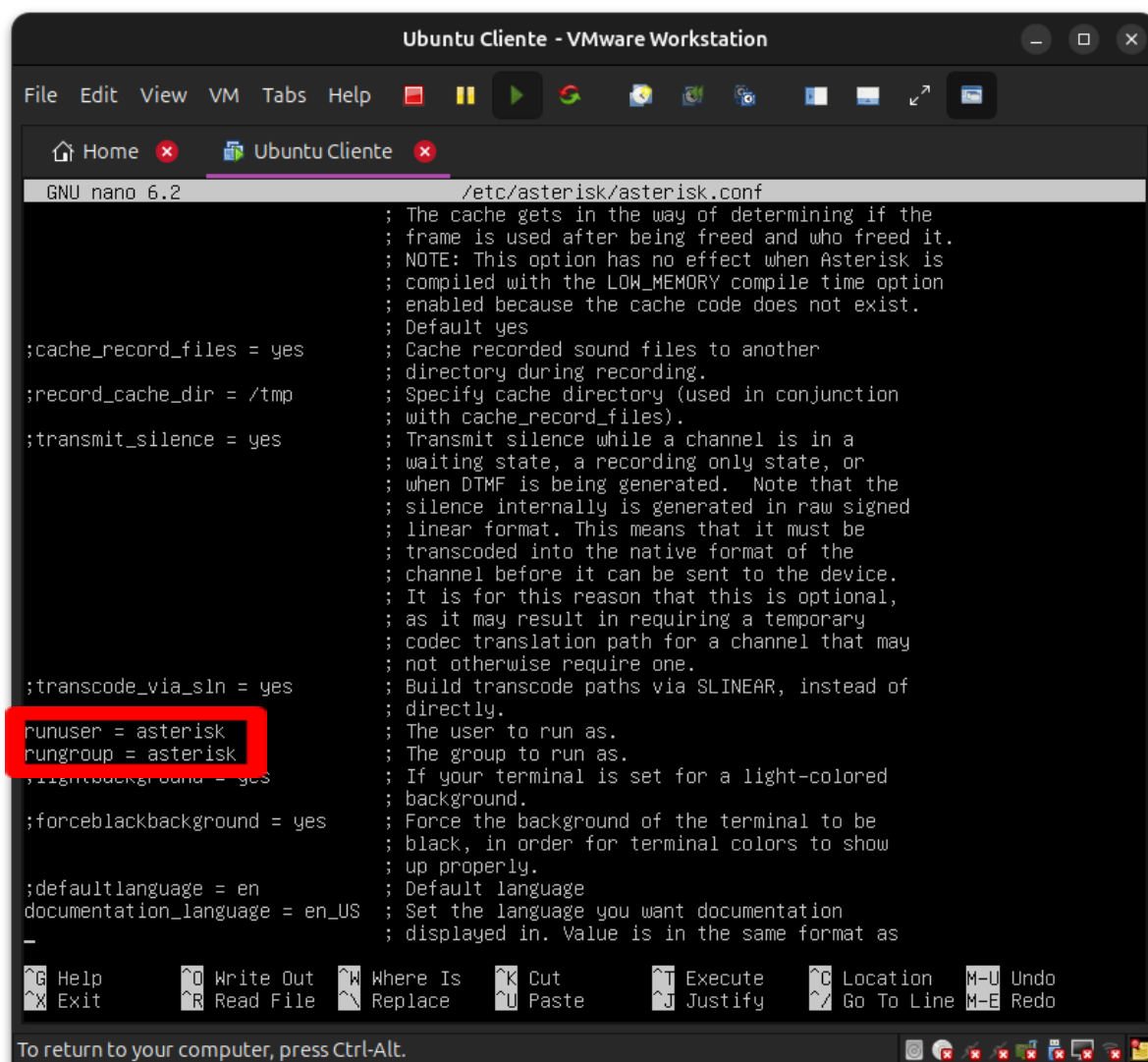
# Asterisk may establish a maximum load average for the system. This
# may be useful to prevent a flood of calls from taking down the system.
#MAXLOAD=4

# Or, if you'd prefer, you can limit the maximum number of calls.
#MAXCALLS=1000

# Default console verbosity. This may be raised or lowered on the console.
# Note this is analogous to the -v command line switch, which by default
# will cause Asterisk to start in console mode and run in the foreground,
# unless the always fork (-F) option is also provided.

[ Read 45 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo

To return to your computer, press Ctrl-Alt.
```



```
GNU nano 6.2 /etc/asterisk/asterisk.conf
; The cache gets in the way of determining if the
; frame is used after being freed and who freed it.
; NOTE: This option has no effect when Asterisk is
; compiled with the LOW_MEMORY compile time option
; enabled because the cache code does not exist.
; Default yes
;cache_record_files = yes      ; Cache recorded sound files to another
;                             ; directory during recording.
;record_cache_dir = /tmp      ; Specify cache directory (used in conjunction
;                             ; with cache_record_files).
;transmit_silence = yes       ; Transmit silence while a channel is in a
;                             ; waiting state, a recording only state, or
;                             ; when DTMF is being generated. Note that the
;                             ; silence internally is generated in raw signed
;                             ; linear format. This means that it must be
;                             ; transcoded into the native format of the
;                             ; channel before it can be sent to the device.
;                             ; It is for this reason that this is optional,
;                             ; as it may result in requiring a temporary
;                             ; codec translation path for a channel that may
;                             ; not otherwise require one.
;transcode_via_sln = yes      ; Build transcode paths via SLINER, instead of
;                             ; directly.
runuser = asterisk            ; The user to run as.
rungroup = asterisk           ; The group to run as.
;lightbackground = yes        ; If your terminal is set for a light-colored
;                             ; background.
;forceblackbackground = yes   ; Force the background of the terminal to be
;                             ; black, in order for terminal colors to show
;                             ; up properly.
;defaultlanguage = en         ; Default language
documentation_language = en_US ; Set the language you want documentation
;                             ; displayed in. Value is in the same format as
```

Ahora editaremos el archivo `/etc/default/asterisk` para establecer el usuario de asterisk por defecto

Descomentamos las siguientes líneas:

AST_USER="asterisk"

AST_GROUP="asterisk"

Editaremos el archivo `/etc/asterisk/asterisk.conf` y **descomentamos** las siguientes líneas:



runuser = asterisk ;The user to run as.

rungroup = asterisk ;The group to run as.

```
ivancli@ivancliente:~/asterisk/src/asterisk-21.12.0$ sudo systemctl status asterisk
• asterisk.service - LSB: Asterisk PBX
   Loaded: loaded (/etc/init.d/asterisk; generated)
   Active: active (running) since Mon 2026-01-26 18:30:59 UTC; 48min ago
     Docs: man:systemd-sysv-generator(8)
    Tasks: 68 (limit: 4512)
   Memory: 50.6M
      CPU: 44.462s
   CGroup: /system.slice/asterisk.service
           └─41257 /usr/sbin/asterisk -U asterisk -G asterisk

ene 26 18:30:59 ivancliente systemd[1]: Starting LSB: Asterisk PBX...
ene 26 18:30:59 ivancliente asterisk[41242]: * Starting Asterisk PBX: asterisk
ene 26 18:30:59 ivancliente asterisk[41242]:   ...done.
ene 26 18:30:59 ivancliente systemd[1]: Started LSB: Asterisk PBX.
ene 26 18:31:10 ivancliente asterisk[41257]: radcli: rc_read_config: rc_read_config: can't open /et>
ene 26 18:31:10 ivancliente asterisk[41257]: radcli: rc_read_config: rc_read_config: can't open /et>
lines 1-16/16 (END)
```

Por último ejecutaremos el **comando sudo systemctl restart asterisk** para que coja las configuraciones correctamente y lo activaremos con el **comando sudo systemctl enable asterisk**



Para terminar ejecutaremos el **comando** `sudo systemctl status asterisk` para comprobar que funciona correctamente, nos tiene que salir **active (running)** en color verde